IT3010
NETWORK DESIGN
MANAGEMENT
TUTORIALS
MID EXAM
REVISION

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Tutorial – 01

1. **Read the case Audi Case Study and drive the information relevant to the five ISO Management Principles.**
   Audi Australia needed a secure, scalable dealer portal for 30 dealerships, around 500 users, and 90 user groups, with role-based access and timely communication

ISO Principle	Audi case එකේ relevant point	Explanation
Fault Management	Old portal was unreliable and dealers sometimes could not access their own documents. Support calls were common. After the new portal, those calls stopped.	මේක network/service faults detect කරලා reduce කරපු example එකක්. Reliable access improve වුණා.
Configuration Management	Audi needed one central portal, permission structures, different user groups, shared publishing rights, in-house administration, and flexible setup. iD also trained admins and structured the portal carefully.	Users, permissions, portal structure, content updates, admin roles manage කිරීම configuration management වලට අයිති.
Accounting Management	Audi used analytic tools to monitor portal activity, track page visits, and track how many forms were lodged per month.	Usage statistics collect කරලා resource use and behavior understand කරන එක accounting management.
Performance Management	Audi wanted timely information, quick access, one-click access to relevant info, and saw 450% user growth and 300% increase in visited pages after launch.	Performance management කියන්නේ system efficient, accessible, responsive තත්ත්වයේ තියාගැනීම.
Security Management	Audi emphasized permission levels, confidential dealership data, and role-based secure access so only the correct users saw relevant information.	Sensitive information protect කරලා authorized usersට විතරක් access දීම security management.

2. **Two areas of network management according to the ISO management framework are Accounting Management and Configuration Management. Compare and contrast these two Network Management processes. Give a detailed explanation about the goals and benefits of each. Explain how these can be used together to help facilitate effective network management.**

Aspect	Configuration Management	Accounting Management
Definition	Network device and system configuration information collect, modify, and store කරන process එක	Network usage statistics gather කරලා resource allocation decisions ගන්න process එක
Main focus	Devices, settings, inventory, configurations	Usage, consumption, quotas, billing
Goal	Network correctly configured and up to date තබාගැනීම	Who uses what resources and how much කියලා track කිරීම
Main activities	Gather info, modify settings, store config data, maintain inventory	Gather usage data, define metrics, set quotas, billing/reporting
Examples	Router config changes, switch VLAN settings, permission structures	User bandwidth usage, number of forms submitted, page visit counts
Benefits	Easy remote config, central storage, better documentation, faster updates	Fair usage, better resource planning, cost control, monitoring user behavior

Configuration management කියන්නේ current network environment ගැන information gather කරලා, ඒ data use කරලා devices modify කරලා, ඒ information store කරලා up-to-date inventory එකක් maintain කරන process එක.
Accounting management කියන්නේ network statistics gather කරලා resource allocation ගැන decisions ගන්න help කරන process එක. එයට usage data gathering, metrics setting, quotas, billing අයිති.

3. **Name the phases of ISO Framework for Network Management.**
   ISO Framework phases/categories are:
   - Fault Management
   - Configuration Management
   - Accounting Management
   - Performance Management
   - Security Management

4. **Briefly explain what is configuration management and its main phases.**
   Configuration management is the process of:
   - gathering information about the current network environment
   - using that data to modify network device configurations
   - storing the data and maintaining inventory/reports

   Main phases:
   - Data Collection – collect information about devices and settings
   - Data Modification – update configuration parameters
   - Data Storing – save configuration data for future use and reporting

5. **Briefly explain the significant advantages of configuration management.**
   - Automatically gathers and updates device data
   - Allows remote configuration of devices
   - Provides central storage for configuration data
   - Helps produce network inventory and reports

6. **Explain two scenarios where configuration management is highly appealing.**
   Scenario 1 – Large organization with many devices
   If a company has many routers, switches, servers, and branches, manually checking every device is slow and error-prone. Configuration management helps auto-discover devices, update settings remotely, and maintain a central inventory. This is especially useful when the network grows large.

   Scenario 2 – Frequent changes in network setup
   If a business often changes routing tables, interfaces, VLANs, services, or user access permissions, configuration management is very useful because it makes modifications easier and keeps configuration records updated. The lecture notes mention routers and servers as examples of modifiable network devices.

7. **What are the possible ways of collecting data for configuration management?**

   Manual collection:
   - done by the administrator manually
   - tedious
   - error-prone
   - time-consuming

Auto-discovery / Auto-mapping

- uses ICMP
- ping
- traceroute
- network management protocol

8. **What are the two main components of an NMS?**

A Network Management System (NMS) comprises:

- An underlying architecture/platform
- A set of applications built on top of the platform

9. **What are the three types of network management architectures?**
    - Centralized
    - Hierarchical
    - Distributed

Tutorial – 02

1. **A network map is a useful tool to a Network Manager, and each layer of the OSI reference model should be documented. Describe at least seven elements that should be documented at the network layer.**
   Network mapping කරනකොට Network Layer (Layer 3) information record කරන්න ඕන.Example elements:
   i. IP addresses of devices
   ii. Subnet masks
   iii. Default gateway
   iv. Routing protocols used (OSPF, RIP etc.)
   v. Router interfaces
   vi. Network topology between routers
   vii. Routing tables
   Network layer documentation =IP addressing + routing information

2. **Describe eight (8) pieces of non-network data that should be collected when mapping a network.**
   Network map එකට non-technical information ද collect කරනවා.
   - Device location
   - Device owner / department
   - Vendor contact details
   - Support contract details
   - Purchase date
   - Warranty information
   - Device administrator contact
   - Maintenance schedule

3. **When should an administrator establish a network baseline?**
   Baseline establish කරන්න ඕන:
   - New network installation time
   - After major upgrades
   - Before troubleshooting
   - During normal network operation

   Baseline = normal network behavior measurement

4. **Discuss three (3) reasons why a network manager should create a baseline for a network.**
   - Detect performance problems
   - Compare normal vs abnormal network behavior
   - Plan network upgrades

   Example:
   - Baseline throughput = 80 Mbps
   - If it suddenly drops to 30 Mbps → problem detected.

5. **You are the newly appointed Network Engineer for ABC Company. The company is in the process of large scale network expansion. Briefly discuss why, when and how you would perform 'Baselining' in your network management process.**

- **Why**
  - ✓ Understand current network performance
  - ✓ Identify bottlenecks
  - ✓ Plan expansion properly
- **When**
  - ✓ Before expansion
  - ✓ After upgrades
  - ✓ During normal traffic periods
- **How**
  - ✓ Monitor network traffic
  - ✓ Measure bandwidth usage
  - ✓ Measure latency
  - ✓ Record utilization statistics

6. **What are the rules/guidelines for creating network documentation?**
   - Use consistent symbols and terminology
   - Define documentation scope clearly
   - Store documents in logical location
   - Update documentation regularly
   - Keep diagrams simple and clear
7. **How long must you monitor to set a network performance baseline?**

   Typical monitoring period: 1–2 weeks

   Reason: *capture normal daily traffic patterns*

   include peak and non-peak hours

8. **"Network Baseline Information Key To Detecting Anomalies." Do you agree to this statement or not? Justify your answer.**
   Network baseline කියන්නේ network එකේ normal performance levels (traffic, bandwidth usage, latency, error rate වගේ) record කරලා තියෙන reference point එකක්. Baseline information තිබුණාම network managerට current network performance එක normal behaviour එකත් එක්ක compare කරන්න පුළුවන්.
   එහෙම compare කරනකොට:
   - unusual traffic spikes
   - high latency
   - packet loss
   - unexpected bandwidth usage

   වගේ anomalies (abnormal behaviour) easily detect කරන්න පුළුවන්.

9. **Which two of the following would be the minimum required and most important components of a network topology diagram?**
   a) Devices
   b) Contact information
   c) Loaded firmware
   d) Connections between devices
   e) Interface spanning tree configuration

10. **Which of the following is not one of the guidelines for creating useful network configuration table must document about each networking device?**
    a) Device name
    b) Data link layer addresses and implemented features
    c) Network layer addresses and implemented features
    d) Size (number of entries) of its routing table
    e) Important information about the physical aspects of the device

11. **Which of the following is not one of the guidelines for creating useful network configuration documentation?**
    a) Use consistent symbols, terminology, and styles
    b) Know the scope of the documentation.
    c) Update the documentation at least once a year
    d) Store the documents in a logical location.

12. List down basic network server types and compare and contrast their advantages and disadvantages.

Server	Function	Advantages	Disadvantages
DNS Server	Domain name resolution	Easy name lookup	Needs maintenance
DHCP Server	Automatic IP assignment	Simplifies network management	Dependency on server
Web Server	Host websites	Central web access	Security risks
File Server	Store shared files	Easy collaboration	Storage management needed

13. What is the Transmission Protocol used for Domain Name Systems? What is the port number? Briefly explain why the protocol used.

- Protocol: UDP
- Port number:53

Why UDP?

- Fast
- Small queries
- Low overhead

TCP used only for:

- large DNS responses
- zone transfers

14. Why only 13 root servers are available in the world?

Historically, DNS protocol design limitation එකක් තිබුණා. Root server list එක single DNS response packet එකකට fit වෙන්න ඕන. එදා UDP packet size limit ~512 bytes. Root server names + IP addresses response එක 512 bytes limit එකට fit වෙන්න පුළුවන් maximum entries ~13 පමණයි. ඒ නිසා 13 root server names (A–M) define කළා.

15. What is the difference between a master server and a slave server with respect to DNS?

Master Server	Slave Server
Primary DNS server	Backup DNS server
Stores original zone data	Copy of master data
Changes made here	Receives updates
Handles zone transfers	Synchronizes with master

16. A student who has logged into the abc.lk domain is trying to request a web page www.webina.aus via Firefox web browser. The domain abc has its own local domain name server and due to security issues in the LAN environment, caching is disabled. Write the steps for resolving www.webina.aus.

- User enters URL in browser
- Browser queries local DNS server
- Local DNS queries root server
- Root server directs to TLD server (.aus)
- TLD server returns authoritative DNS server
- Authoritative server returns IP address
- Browser connects to web server

17. What type of query methods are used in the following servers?

a) Root domain name server: **Iterative**
b) Local domain name server: **Recursive**
c) Top level domain name server: **Iterative**
d) Authoritative domain name server: **Iterative**

18. Why cached data is periodically refreshed or removed?

Reasons:

- Data becomes outdated
- DNS changes occur
- Ensure accuracy
- Avoid incorrect responses

19. What is the Transmission Protocol used in a DHCP server? What are the server side and clientside port numbers used? Briefly explain.

Protocol: UDP. **Why UDP?** Fast broadcast communication.

Side	Port
Client	68
Server	67

20. DHCP messages are always uses the broadcasting method in communication. Why? Client does not know:

- its IP address
- DHCP server address

Broadcast helps client discover available DHCP servers.

21. Briefly explain the LAMP Web Service Solution Stack?

Component	Meaning
L	Linux
A	Apache
M	MySQL
P	PHP

22. Why is the name PROXY used for the proxy server?

Proxy means an intermediate representative.

Proxy server acts as: **User → Proxy → Internet**

Benefits:

- security
- caching
- anonymity

23. Caching is one of the main features in a proxy server. What is the purpose of using a cache?

Cache stores frequently accessed content.

Benefits:

- faster responses
- reduced internet bandwidth
- improved performance

24. What is the purpose of the following cache types?

Cache Type	Purpose
Browser Cache	Store web pages locally for faster loading
Proxy Cache	Store frequently accessed internet resources
Server Cache	Store processed data to reduce server workload

Tutorial – 03

1. **List down basic network server types and compare and contrast their advantages and disadvantages.**

Server Type	Function	Advantages	Disadvantages
DNS Server	Domain names → IP addresses convert කරනවා	Easy access to websites	Needs maintenance
DHCP Server	Automatic IP address assignment	Simplifies network setup	Server failure affects clients
Web Server	Host websites and web applications	Central web content delivery	Security risks
File Server	Shared file storage	Easy collaboration	Storage management required
Mail Server	Email services	Centralized communication	Spam/security issues

2. **What is the Transmission Protocol used for Domain Name Systems? What is the port number? Briefly explain why the protocol used.**
   - Protocol used: UDP
   - Port number: 53

   Why UDP? Reasons:
   - Faster communication
   - Small request/response messages
   - Low overhead

   When TCP used?
   - Zone transfers
   - Large DNS responses

3. **Why only 13 root servers are available in the world?**

   Reasons:
   - Historical DNS design limitation
   - DNS packet size restrictions
   - Efficient global query routing

   Each root server has many distributed mirror instances worldwide using Anycast. So actually hundreds of servers exist globally.

4. **What is the difference between a master server and a slave server with respect to DNS?**

Master Server	Slave Server
Primary DNS server	Secondary DNS server
Stores original zone data	Stores replicated copy
Administrators update records here	Updates from master via zone transfer
Controls DNS records	Provides redundancy

Purpose: Improve **DNS reliability and load balancing**.

5. **A student who has logged into the abc.lk domain is trying to request a web page www.webina.aus via Firefox web browser. The domain abc has its own local domain name server and due to security issues in the LAN environment, caching is disabled. Write the steps for resolving www.webina.aus.**
Steps:

1. User types www.webina.aus in browser
2. Browser sends request to Local DNS server (abc.lk)
3. Local DNS queries Root DNS server
4. Root server returns TLD server (.aus)
5. Local DNS queries TLD server
6. TLD server returns Authoritative DNS server
7. Local DNS queries Authoritative DNS server
8. Authoritative server returns IP address
9. Browser connects to web server

Flow Diagram (concept)
User - Local DNS - Root DNS - TLD DNS (.aus) - Authoritative DNS - IP returned - Browser connects to web server

6. **What type of query methods are used in the following servers?**
   a) Root domain name server: **Iterative**
   b) Local domain name server: **Recursive**
   c) Top-level domain name server: **Iterative**
   d) Authoritative domain name server: **Iterative**

7. **Why cached data is periodically refreshed or removed?**

   Reasons:

   - DNS records change
   - Prevent outdated information
   - Maintain accuracy
   - Improve network efficiency

   TTL (Time To Live) controls cache expiry.

8. **What is the Transmission Protocol used in a DHCP server? What are the server-side and client-side port numbers used? Briefly explain.**

   Protocol: UDP

   Port Numbers:

Device	Port
DHCP Server	67
DHCP Client	68

   Reason: UDP allows fast broadcast communication.

9. **DHCP messages are always used in the broadcasting method in communication. Why?**

   When the client first joins the network, the client does NOT know:

   - its IP address
   - DHCP server address

   Therefore, it sends a broadcast: DHCPDISCOVER

   Server replies with: DHCPOFFER

10. **Briefly explain the LAMP Web Service Solution Stack?**

Letter	Technology
L	Linux
A	Apache
M	MySQL
P	PHP

   Purpose: Build dynamic web applications.

   Example: WordPress server.

11. Why the name PROXY is used for the proxy server?

Proxy = intermediate representative

Architecture: Client → Proxy Server → Internet

A proxy acts as an intermediary between the user and external servers.

Benefits:

- Security
- Caching
- Content filtering

12. Caching is one of the main features of a proxy server. What is the purpose of using a cache?

Proxy cache stores frequently requested data.

Benefits:

- Faster access
- Reduced internet bandwidth
- Lower server load

Example:

If 100 employees visit same website → cache serves it once downloaded.

13. What is the purpose of the following cache types?

Cache Type	Purpose
Browser Cache	Stores web pages locally on user device
Proxy Cache	Stores frequently accessed internet content
Server Cache	Stores processed data on server for faster responses

Tutorial – 04

1. Briefly describe NMS-Agent architecture. Also include in your discussion what MDB is.

SNMP network management system consists of three main components:

- **NMS (Network Management System / Manager)**
- **Agent**
- **MIB Database**

Architecture
Network Manager (NMS) - SNMP Request – Agent - MIB Database

NMS (Manager)
Central management system that monitors and controls network devices.

Agent
Software running inside network devices (router, switch, server) that collects device information.

MIB (Management Information Base)
Database that stores information about network devices.

MDB (Management Database)
MDB contains management information collected from multiple network devices and is used by the NMS for monitoring and analysis.

2. How (history) and why (goals) was SNMP initially introduced?

SNMP was introduced in the late 1980s when computer networks rapidly expanded and became difficult to manage manually.

Goals of SNMP

- Provide simple network monitoring
- Manage network devices remotely
- Detect faults in network devices
- Collect performance data
- Enable centralized network management

SNMP was designed to be lightweight and easy to implement in different networking devices.

3. What are the four models defined in SNMPv1 architecture?

SNMPv1 architecture defines four models:

Model	Description
Organization Model	Defines components of SNMP system
Information Model	Defines managed objects in MIB
Communication Model	Defines message exchange process
Functional Model	Defines SNMP operations

4. What are the two entities in SNMPv1? Also indicate the main purpose of each entity.

Two entities in SNMPv1:

Entity	Purpose
Manager (NMS)	Controls and monitors network devices
Agent	Provides device information to manager

Manager sends requests and agent responds with data.

5. What is MIB? And what is the method used to identify objects within the MIB uniquely?

MIB (Management Information Base) is a database that stores information about network devices such as performance, status, and configuration.

Objects inside the MIB are uniquely identified using an **OID (Object Identifier).**

Example: 1.3.6.1.2.1 (Each number represents a node in the MIB hierarchy.)

6. What is the language used to define the objects in the MIB?

Objects in the MIB are defined using: **ASN.1 (Abstract Syntax Notation One)**

ASN.1 is a formal language used to define data structures for network communication.

7. What are the characteristics of SNMPv1 message exchange?

SNMPv1 message exchange has the following characteristics:

- Uses UDP protocol
- Simple request–response communication
- Manager initiates communication
- Lightweight protocol
- Uses community strings for authentication

8. Briefly explain the SNMPv1 administrative model.

- The SNMPv1 administrative model defines how access control and authentication are handled.
- It uses **community strings** to control access to network devices.
- Community strings act like passwords that determine whether a user can read or modify management data.

9. What are the purposes and disadvantages of using community strings in SNMPv1? What is the default community string?

Community strings provide basic authentication between the SNMP manager and agent.

Disadvantages

- Sent as plain text
- Weak security
- Can be intercepted by attackers
- No encryption

Default community string: public (Read-only access)

10. List the five types of operations defined in SNMPv1.

Five SNMPv1 operations:

1. **GET** – Retrieve the value of an object
2. **GETNEXT** – Retrieve next object in MIB tree
3. **SET** – Modify value of an object
4. **RESPONSE** – Reply from agent
5. **TRAP** – Alert message from agent

11. What are the main steps of sending a request in SNMPv1? Who is performing these steps?

Performed by the **Manager (NMS)**.

Steps:

1. Create SNMP request message
2. Add community string
3. Specify object identifier (OID)
4. Send request to the agent

12. What are the main steps of receiving a request in SNMPv1? Who is performing these steps?

Performed by the **Agent**.

Steps:

1. Receive SNMP request
2. Verify community string
3. Locate requested object in MIB
4. Process the request

13. What are the main steps of sending a response in SNMPv1? Who is performing these steps?

Performed by the **Agent**.

Steps:

1. Retrieve requested value from MIB
2. Create SNMP response message
3. Send response back to manager

14. What are the main steps of receiving the response in SNMPv1? Who is performing these steps?

Performed by the **Manager (NMS)**.

Steps:

1. Receive response message
2. Extract returned data
3. Display or store information

15. Briefly explain how SNMPv1 GET Request is processed.

GET request retrieves the value of a specific MIB object.

Manager - GET Request (OID) – Agent - Search MIB - Response returned

Example: retrieving CPU usage of a router.

16. What are the three errors that you may receive for a GET Request operation in SNMPv1?

Three possible errors:

Error	Meaning
noSuchName	Requested object not found
tooBig	Response message too large
genErr	General error

17. Briefly explain how the SNMPv1 GETNEXT Request is processed.

GETNEXT retrieves the **next object in the MIB hierarchy.**

- Manager → GETNEXT request
- Agent → next OID returned

Used to browse the MIB structure.

18. Explain how GET and GETNEXT together retrieve the entire MIB tree.

Process:

1. Manager sends **GET request** for first object
2. Agent returns object value
3. Manager repeatedly sends **GETNEXT requests**
4. Each request returns next object in tree

Example: OID1 → OID2 → OID3 → OID4
Manager walks through the entire MIB tree.

19. Briefly explain how SNMPv1 SET Request is processed.

SET request modifies the value of an object.

Manager - SET Request – Agent - Modify MIB value - Response returned

Example: changing router configuration.

20. What are the three errors that you may receive for a SET Request operation in SNMPv1?

Possible errors:

Error	Meaning
noSuchName	Object does not exist
badValue	Invalid value
genErr	General error

21. What is the special condition that should be satisfied when SET Request is processed?

All variables must be valid before updating. If one variable fails → entire SET request fails.

This is called an **Atomic Operation**.

22. What is the usage of Trap messages in SNMPv1?

Trap messages notify the manager about important events.

Examples:

- device reboot
- link failure
- authentication error

Trap is sent **without manager request**.

23. What are the 7 types of Trap messages defined in SNMPv1?

1. coldStart
2. warmStart
3. linkDown
4. linkUp
5. authenticationFailure
6. egpNeighborLoss
7. enterpriseSpecific

24. Explain the limitations of SNMPv1.

Limitations:

- Weak security
- No encryption
- Plain-text authentication
- Limited error handling
- No strong access control

These problems were improved in SNMPv2 and SNMPv3.

25. Compare SNMPv1 Gauge and Counter data types.

Feature	Counter	Gauge
Definition	Value that increases continuously	Value that can increase or decrease
Example	Number of packets sent	CPU utilization
Reset behavior	Resets after maximum value	Changes dynamically

Tutorial – 06

1. What are the 11 separate groups defined in MIB-II?

The 11 groups defined in **MIB-II** are:

1. System
2. Interfaces
3. Address Translation
4. IP
5. ICMP
6. TCP
7. UDP
8. EGP
9. Transmission
10. SNMP
11. CMOT

2. Briefly explain the purpose of each above group including some objects.

MIB Group	Purpose	Example Objects
System	General device information	sysName, sysLocation, sysContact
Interfaces	Network interface information	ifNumber, ifSpeed
Address Translation	IP-to-MAC address mapping	atTable
IP	IP protocol statistics	ipInReceives
ICMP	ICMP protocol statistics	icmpInMsgs
TCP	TCP connection information	tcpConnTable
UDP	UDP statistics	udpInDatagrams
EGP	Exterior Gateway Protocol data	egpInMsgs
Transmission	Media-specific information	Ethernet statistics
SNMP	SNMP protocol statistics	snmpInPkts
CMOT	OSI protocol related management	cmotObjects

3. What are the major changes/additions included in SNMPv2?

Major improvements in SNMPv2 include:

- Introduction of **GETBULK operation**
- Improved **error handling**
- **Two-phase commit architecture** for SET requests
- Manager-to-manager communication using **INFORM**
- Additional performance monitoring capabilities

These improvements made SNMPv2 more efficient than SNMPv1.

4. What is the major change to GET and GETNEXT processing in SNMPv2?

The major enhancement was the introduction of **GETBULK operation**. GETBULK allows retrieving **large amounts of MIB data in one request**, reducing the number of requests needed. This improves **network management efficiency and reduces traffic**.

5. What is the two-phase commit architecture of SET request processing in SNMPv2?

SET request processing in SNMPv2 follows a **two-phase commit process**.

Phase 1 – Validation Phase

- The agent checks whether all variables are valid.

Phase 2 – Commit Phase

- If validation succeeds, all variables are updated.

Benefit

Ensures **data consistency** and prevents partial updates. If any variable fails → no changes are made.

6. Briefly explain how GETBULK Request operates.

GETBULK retrieves **large blocks of data from the MIB** in a single request.

Steps:

1. Manager sends GETBULK request
2. Agent retrieves multiple objects from MIB
3. Agent sends a large response containing multiple values

Example: Manager → GETBULK

Agent → Multiple MIB values returned

Benefit: reduces the number of requests required.

7. What are the limitations of GETBULK Request in SNMPv2?

Limitations include:

- Large responses may overload the network
- Increased processing load on agent
- May exceed maximum UDP packet size
- Difficult to control exact number of objects returned

8. What is INFORM Request operation in SNMPv2 and why is it useful?

INFORM request allows **manager-to-manager communication**. Unlike Trap messages, INFORM requires **acknowledgment** from the receiver.

Benefit: Ensures that the notification was successfully delivered.

9. What are the main features introduced in SNMPv3?

SNMPv3 introduced major security improvements:

- Strong authentication
- Encryption for privacy
- Message integrity verification
- Access control mechanisms

SNMPv3 solved many security issues of SNMPv1 and SNMPv2.

10. What are the three main security features SNMPv3 protects against?

SNMPv3 protects against:

1. **Modification of information**
2. **Masquerade attacks (fake identity)**
3. **Disclosure of information (data interception)**

These are handled using authentication and encryption.

11. What are the two main parts of a SNMPv3 entity?

An SNMPv3 entity consists of two main parts:

1. **SNMP Engine**
2. **SNMP Applications**

12. Explain the four parts of SNMPv3 engine.

Component	Purpose
Dispatcher	Sends and receives messages
Message Processing Subsystem	Processes SNMP messages
Security Subsystem	Provides authentication and encryption
Access Control Subsystem	Controls user permissions

13. What are the five types of applications in SNMPv3 entity?

Application	Purpose
Command Generator	Sends SNMP requests
Command Responder	Responds to requests
Notification Originator	Generates trap/inform messages
Notification Receiver	Receives trap/inform messages
Proxy Forwarder	Forwards SNMP messages

14. Illustrate SNMPv3 architecture.

Example architecture:

```
SNMP Applications
   │
   │
SNMP Engine
 ├ Dispatcher
 ├ Message Processing
 ├ Security Subsystem
 └ Access Control
```

Applications interact with the SNMP engine which manages security, message handling, and communication.